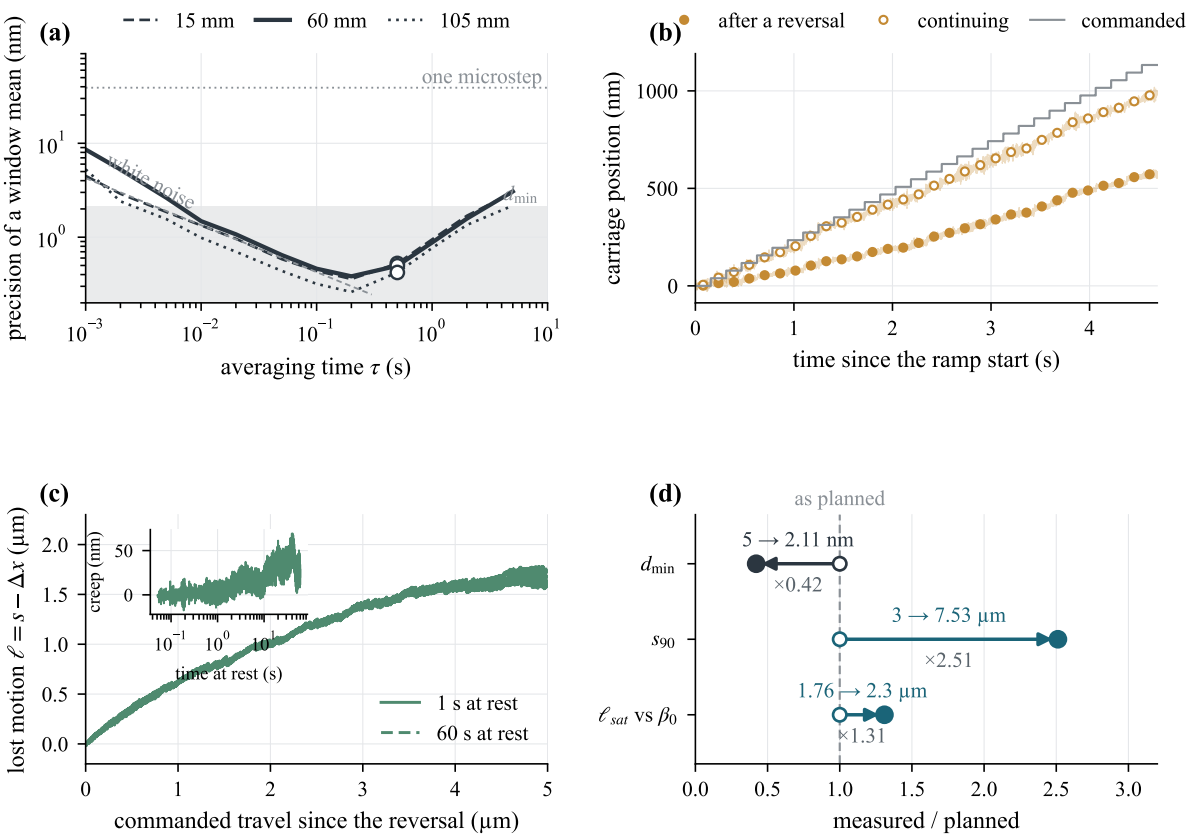

(a)
15 mm
60 mm
105 mm
one microstep
white noise
σ_{min}
precision of a window mean (nm)
averaging time τ (s)
(b)
after a reversal
continuing
commanded
carriage position (nm)
time since the ramp start (s)
(c)
lost motion $\ell = s - \Delta x$ (µm)
creep (nm)
time at rest (s)
1 s at rest
60 s at rest
commanded travel since the reversal (µm)
(d)
as planned
d_{min}
5 → 2.11 nm
×0.42
s_{90}
3 → 7.53 µm
×2.51
ℓ_{sat} vs β_0
1.76 → 2.3 µm
×1.31
measured / planned